

# Kanban Board — Project Plan

Vanilla JS · Drag & Drop · localStorage — Single Board (v1)

## Domain

A workflow/task management tool. Items ('cards') move through stages ('columns') left to right as they progress. The column a card sits in IS its status — there is no separate status field. Generic engine: user can name/add/remove columns rather than columns being hardcoded to one specific use case.

## Data Structure — Flat (not nested)

Cards are stored in a single flat array, each pointing to its column via **columnId**. Columns do NOT hold their cards inside themselves. Moving a card between columns is a one-line update (`card.columnId = newId`) instead of removing/pushing between nested arrays.

```
columns = [ { id, name, color } ]  
cards   = [ { id, title, description, columnId } ]
```

## Classes (v1 — no Board entity)

```
class Column { constructor(id, name, color) }  
class Card { constructor(id, title, description, columnId) }
```

Board entity intentionally dropped for v1 to keep scope small. If added later (multi-board / sidebar), just add `boardId` back onto Column — Card needs no changes.

## Features — v1 (Core)

- Columns: default set on load (To Do / In Progress / Done); user can add, rename, delete columns
- Create card — add a new card into a specific column (title required, description optional)
- Edit card — click a card, modal opens (reuse SweetAlert2 pattern), edit title/description, save
- Delete card — with confirm dialog (reuse SweetAlert2 confirm pattern)
- Drag & drop cards — move a card between columns, updates `card.columnId`
- Persist to localStorage — save on every change, load on page refresh
- Column color-coding — card visually styled by its column's color (color lives on Column, not Card)

## Features — Optional / Nice-to-have

- Card count per column (e.g. 'To Do (4)')
- Empty state message when a column has zero cards
- Delete column — decide behavior for cards still inside it
- Due dates / labels on cards

## Explicitly Out of Scope for v1

- Sidebar / multiple boards / active board switching
- Statistics view (Chart.js)
- In-column card reordering (would need an 'order' field — deferred)
- Any API calls, backend, or authentication

Note: multi-board/sidebar is planned as a future v2, once React is learned — at that point boardId returns to Column and a sidebar drives an activeBoardId, same filtering pattern used throughout v1.

## Next Step

Class diagram → activity diagram → flowchart → README → git init → code (HTML → CSS → JS), committing per feature.